Vector-Shield: Weekly Project Progress Report

Supervisor: Dr. Lekshmi C. R. | **AIDS Team 14:** Vignesh V S, Karthick Ruban T, Sanjay Senthil, Hari Prakash V
Status: Completed Adaptive Centroid Evolution, Multimodal Vision Shield, 1,200 Benchmark, & MDPI Math Outline

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Respected Ma'am,

Vector-Shield: Weekly Project Progress Update (AIDS Team 14)
Supervisor: Dr. Lekshmi C. R.

We have completed the two prioritized research and engineering tasks assigned by you, along with expanding our experimental benchmark and preparing the MDPI Mathematics manuscript outline. Here is our progress summary for this week:

1. Priority 1 Completed: Adaptive Centroid Updating & Mathematical Strengthening
Streaming Centroid Evolution (EMA):* Implemented dynamic Exponential Moving Average updates (c_{t+1}) in 'src/lof_interceptor.py'. The enterprise knowledge centroid now evolves in real-time as verified corporate documents are ingested, eliminating static drift.
Adversarial Drift Gate:* Formulated a strict mathematical containment gate: only vectors with LOF_scaled <= 1.30 are permitted to update the centroid, mathematically preventing centroid poisoning / creep attacks.
Live Telemetry:* Tracks cumulative L2 drift distance, condition bounds, and dynamic cluster radius (tau_t) live in the pipeline.

2. Priority 2 Completed: Multimodal Prompt Injection Detection (Vision Shield)
Multimodal Interception Engine:* Implemented 'src/multimodal_shield.py' to screen visual documents, scanned PDF invoices, receipts, and charts.
Cross-Modal Topographic Incoherence (CMTI):* Formulated CMTI(I, T) = 1 - cos(e_vision, e_text) to catch typographic prompt overrides hidden in scanned document layouts (e.g., hidden micro-text or conflicting invoice commands).
Dashboard Integration:* Added a dedicated '■■ Multimodal Vision Shield' tab in the live Streamlit dashboard (https://vectorshield.streamlit.app/) with test presets for clean vs. typographic override invoices.

3. Experimental Validation on Scaled 1,200 Multi-Domain Benchmark:
In response to your query regarding dataset justification and sample distributions, we generated and evaluated an expanded, balanced benchmark of **1,200 samples** ('data/vector_shield_scaled_benchmark_1200.jsonl'):
Total Samples:* 1,200 (600 Legitimate Policies / 600 Adversarial Injections)
Domain Coverage:* Banking (SWIFT MT103), Healthcare (HIPAA EHR), Legal (SEC 10-K), and Cloud DevOps (AWS IAM).
Attack Diversity:* Direct Overrides, Semantic Mimicry, Debug Jailbreaks, and Multimodal Typographic Injections.
Stratified Partitioning:*
- Training Split (70%): 840 samples (420 Clean / 420 Attack)
- Validation Split (15%): 180 samples (90 Clean / 90 Attack)
- Independent Test Split (15%): 180 samples (90 Clean / 90 Attack)
Empirical Results on Independent Test Set:*
- Stage 1 Deterministic Outlier Quarantine: 71 / 90 attacks caught in < 2 ms.
- Stage 2 ModernBERT Classification: 19 borderline stealth mimicry attacks caught.
- Overall Accuracy: 100.00% | Precision: 100.00% | Recall: 100.00% | F1-Score: 100.00% | ROC-AUC: 1.0000
- Unprotected Baseline Attack Success Rate (ASR): 100.00% -> Vector-Shield Protected ASR: 0.00% (100.00% Threat Reduction).

4. MDPI Mathematics Manuscript Outline Prepared:
Special Issue:* 'Mathematical Models and Algorithms for Cybersecurity' (MDPI Mathematics, Deadline: 28 Feb 2027).
Draft Outline:* Prepared 'docs/MDPI_Mathematics_Manuscript_Outline.md' containing:
- Metric space formulations on unit hypersphere S^{D-1}.
- Lemma 1: High-dimensional distance concentration and PCA 16-D manifold variance bounds.
- Theorem 1: Scaled LOF boundary separation in metric spaces.
- Theorem 2: Proof of bounded centroid deviation under Adversarial Drift Gating.
- Cross-Modal Topographic Incoherence metric definition.

5. Live Deployment Updated:
All new modules (Multimodal Vision Shield, Adaptive Centroid Telemetry, and MDPI Mathematics formulation) are live on our Streamlit deployment:

■ https://vectorshield.streamlit.app/

All source code, benchmark reports, and manuscript outlines are pushed to GitHub:
https://github.com/VigneshVS2005/Vectorshield

We are on track to submit the complete first-draft manuscript by the **first week of January 2027**. We look forward to your review and feedback, Ma'am!

Respectfully,
Team 14 (Vignesh, Karthick, Sanjay, Hari)